



CSC104-Database Systems Mid Project Report(Spring 2025)



Student Details

Repository Name	DBS25P048
Registration Number	2024-CS-48

Project	
Project Title	ITEC Event Management System
Executive Summary	<i>The ITEC Event Management System is a centralized desktop application developed to streamline the organization of annual ITEC events at the Department of Computer Science, UET Lahore. The system replaces manual processes involving spreadsheets and physical records, addressing critical challenges such as:</i> <i>Data Inconsistency: Automates participant registration, committee assignments, and financial tracking.</i> <i>Efficiency: Provides tools for scheduling events, allocating venues, and managing duties.</i> <i>Reporting: Generates real-time reports for participants, finances, and event schedules.</i> <i>Built using C# WinForms and MySQL, the application follows a 3-tier architecture (UI, BL, DL) with direct SQL queries for database operations (no Entity Framework). Key features include:</i> <i>Multi-edition event management</i> <i>Dynamic financial tracking (income/expenses)</i> <i>Conflict-free venue allocation</i> <i>Automated report generation</i> <i>The system adheres to strict schema guidelines and uses a normalized database with 15+ tables (committees, participants, finances, etc.).</i>

ITEC

Dashboard
Events
Participants
Venues
Committees
Finances
Duties
Editions
Report
Logout

Event Management

Events Home
Event Participants
Event Results

Event Name:
Year:
Event Category:
Committee:
Venue:
Event Date:
Description:

Insert
Delete
Update

	Year	Event Name	Category	Description	Event Date	Venue	Committee
▶	2023	sd	Seminar	jn	2023-07-24	Main Auditori...	Logistics Com...
	2024	sdf	Competition	jsnda	2023-07-24	Seminar Hall	Technical Com...
	2023	AI Hackathon	Competition	asdfa	2023-03-15	Main Auditori...	Technical Com...
	2023	Tech Summit	Seminar	asdfsdf	2023-03-18	Seminar Hall	Logistics Com...
	2024	new	Exhibition	jas	2025-07-24	Seminar Hall	Technical Com...

UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
cmbItec	Combo Box	EventsManageUC.cs	SELECT itec_id, year FROM itec_editions
cmbCategory	Combo Box	EventsManageUC.cs	SELECT event_category_id, category_name FROM event_categories
cmbVenue	Combo Box	EventsManageUC.cs	SELECT venue_id, venue_name FROM venues
UpdateBtn	Button	EventsManageUC.cs	<pre> \$@"UPDATE itec_events SET itec_id={updatedEvent.ItecId}, event_name ='{updatedEvent.EventName}', event_category_id ={updatedEvent.EventCategoryId}, description ='{updatedEvent.Description}', event_date ='{updatedEvent.EventDate:yyyy-MM-dd}', venue_id={updatedEvent.VenueId}, committee_id ={updatedEvent.CommitteeId} WHERE event_id = {updatedEvent.EventId}" </pre>
cmbCommittee	Combo Box	EventsManageUC.cs	SELECT committee_id, committee_name FROM committees

AddBtn	Button	EventsManageUC.cs	\$@"INSERT INTO itec_events (itec_id, event_name, event_category_id, description, event_date, venue_id, committee_id) VALUES ({newEvent.ItecId}, '{newEvent.EventName}', {newEvent.EventCategoryId}, '{newEvent.Description}', '{newEvent.EventDate:yyyy-MM-dd}', {newEvent.VenueId}, {newEvent.CommitteeId})"
DataGridView1	DataGridView	EventsManageUC.cs	@"SELECT e.event_id, e.itec_id, e.event_name, e.event_category_id,e.description,e.eve nt_date, e.venue_id,e.committee_id,ie.year as itec_year, ec.category_name as event_category_name, v.venue_name,c.committee_name FROM itec_events e LEFT JOIN itec_editions ie on e.itec_id = ie.itec_id LEFT JOIN event_categories ec on e.event_category_id = ec.event_category_id LEFT JOIN venues v on e.venue_id = v.venue_id LEFT JOIN committees c on e.committee_id =c.committee_id"
DeleteBtn	Button	EventsManageUC.cs	\$"DELETE FROM itec_events WHERE event_id = {eventId}"

Interface Name		EventsParticipantUC															
Event Management Events Home Event Participants Event Results Event : AI Hackathon Participant: shahzaib Status: Pending Amount: 0 Insert Update Delete <table border="1"> <thead> <tr> <th></th> <th>EventName</th> <th>ParticipantName</th> <th>PaymentStatus</th> <th>FeeAmount</th> </tr> </thead> <tbody> <tr> <td>▶</td> <td>Tech Summit</td> <td>shahzaib</td> <td>Paid</td> <td>Rs100.00</td> </tr> <tr> <td>*</td> <td></td> <td></td> <td></td> <td></td> </tr> </tbody> </table>				EventName	ParticipantName	PaymentStatus	FeeAmount	▶	Tech Summit	shahzaib	Paid	Rs100.00	*				
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▶	Tech Summit	shahzaib	Paid	Rs100.00													
*																	

UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
AddBtn	Button	EventsParticipantsUC.cs	\$@" INSERT INTO event_participants (event_id, participant_id, payment_status_id, fee_amount) VALUES ({participant.EventId}, {participant.ParticipantId}, {participant.PaymentStatusId}, {participant.FeeAmount})"
UpdateBtn	Button	EventsParticipantsUC.cs	\$@" UPDATE event_participants SET event_id = {participant.EventId}, participant_id = {participant.ParticipantId}, payment_status_id = {participant.PaymentStatusId}, fee_amount = {participant.FeeAmount} WHERE registration_id = {participant.RegistrationId}"
DeleteBtn	Button	EventsParticipantsUC.cs	\$"DELETE FROM event_participants WHERE registration_id = {registrationId}"
cmbEvent	Combo Box	EventsParticipantsUC.cs	"SELECT event_id, event_name FROM itec_events"
cmbParticipant	Combo Box	EventsParticipantsUC.cs	"SELECT participant_id, name FROM participants"
cmbStatus	Combo Box	EventsParticipantsUC.cs	"SELECT lookup_id, value FROM lookup WHERE category = 'PaymentStatus'"
DataGridView1	DataGridView	EventsParticipantsUC.cs	@ " SELECT ep.registration_id, ep.event_id, ep.participant_id, ep.payment_status_id, ep.fee_amount, e.event_name, p.name AS participant_name, l.value as payment_status FROM event_participants ep LEFT JOIN itec_events e on ep.event_id = e.event_id LEFT JOIN participants p on ep.participant_id = p.participant_id LEFT JOIN lookup l on ep.payment_status_id = l.lookup_id"

Event Management						
Events Home		Event Participants		Event Results		
Event Name:	AI Hackathon	Participant:	shahzaib	Team:		
position:	1	Score:	90	Remarks: ajkd		
Insert		Update		Delete		

Event	ParticipantName	TeamName	Position	Score	Remarks
new	lasnd	al	1	90.00	uqd
new	asdj	asasjdkn	2	89.00	asjhd
*					

UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
AddBtn	Button	EventsResultsUC.cs	\$@"INSERT INTO event_results (event_id, participant_id, team_id, position, score, remarks) VALUES ({result.EventId}, {participantPart}, {teamPart}, {result.Position}, {result.Score}, '{result.Remarks}')
UpdateBtn	Button	EventsResultsUC.cs	\$@"UPDATE event_results SET event_id = {result.EventId}, participant_id = {participantPart}, team_id = {teamPart}, position = {result.Position}, score = {result.Score}, remarks = '{result.Remarks}' WHERE result_id = {result.ResultId}"
DeleteBtn	Button	EventsResultsUC.cs	\$"DELETE FROM event_results WHERE result_id = {resultId}"
cmbEvent	Combo Box	EventsResultsUC.cs	"SELECT event_id, event_name FROM itec_events"
cmbParticipant	Combo Box	EventsResultsUC.cs	"SELECT participant_id, name FROM participants"
cmbTeam	Combo Box	EventsResultsUC.cs	"SELECT team_id, team_name FROM teams"

DataGridView1	DataGridView	EventsResultsUC.cs	@ "SELECT er.result_id, er.event_id, er.participant_id, er.team_id, er.position, er.score, er.remarks, e.event_name, p.name AS participant_name, t.team_name FROM event_results er LEFT JOIN itec_events e on er.event_id = e.event_id LEFT JOIN participants p on er.participant_id = p.participant_id LEFT JOIN teams t on er.team_id = t.team_id"
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Interface Name		ParticipantsUC																											
Event Management Events HomeEvent ParticipantsEvent Results Event Name:Al Hackathon Participant:shahzaib Team: position:1 Score:90 Remarks:ajkd Insert Delete Update <table><thead><tr><th>Event</th><th>ParticipantName</th><th>TeamName</th><th>Position</th><th>Score</th><th>Remarks</th></tr></thead><tbody><tr><td>new</td><td>lasnd</td><td>al</td><td>1</td><td>90.00</td><td>uqd</td></tr><tr><td>new</td><td>asdj</td><td>asasjdkn</td><td>2</td><td>89.00</td><td>asjhd</td></tr><tr><td>*</td><td></td><td></td><td></td><td></td><td></td></tr></tbody></table>						Event	ParticipantName	TeamName	Position	Score	Remarks	new	lasnd	al	1	90.00	uqd	new	asdj	asasjdkn	2	89.00	asjhd	*					
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new	asdj	asasjdkn	2	89.00	asjhd																								
*																													
UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table																										
AddBtn	Button	ParticipantsUC.cs	\$@"INSERT INTO participants (itec_id, name, email, contact, institute, role_id) VALUES ({participant.ItecId}, '{participant.Name}', '{participant.Email}', '{participant.Contact}', '{participant.Institute}', {participant.RoleId})"																										
UpdateBtn	Button	ParticipantsUC.cs	\$@"UPDATE participants SET itec_id = {participant.ItecId},																										

			name = '{participant.Name}', email = '{participant.Email}', contact = '{participant.Contact}', institute = '{participant.Institute}', role_id = {participant.RoleId} WHERE participant_id = {participant.ParticipantId}"
DeleteBtn	Button	ParticipantsUC.cs	\$"DELETE FROM participants WHERE participant_id = {participantId}"
cmbItec	Combo box	ParticipantsUC.cs	"SELECT itec_id, year FROM itec_editions"
cmbRole	Combo box	ParticipantsUC.cs	"SELECT role_id, role_name FROM roles"
DataGridView1	DataGridView	ParticipantsUC.cs	"SELECT p.participant_id, p.name, p.itec_id, ie.year, p.email, p.contact, p.institute, p.role_id, r.role_name as r_name FROM participants p LEFT JOIN roles r on p.role_id=r.role_id LEFT JOIN itec_editions ie on p.itec_id=ie.itec_id "

<i>Interface Name</i>	VenuesUC																				
Venues Venues Venue Allocations Venue Name: Capacity: Location: Insert Update Delete Search <table border="1" style="width: 100%; border-collapse: collapse; text-align: center;"> <thead> <tr style="background-color: #d9d9d9;"> <th style="width: 5%;"></th> <th style="width: 35%;">Venue Name</th> <th style="width: 20%;">Capacity</th> <th style="width: 40%;">Location</th> </tr> </thead> <tbody> <tr> <td>▶</td> <td>Main Auditorium</td> <td>1000</td> <td>Block A</td> </tr> <tr> <td></td> <td>Seminar Hall</td> <td>300</td> <td>Block B</td> </tr> <tr> <td></td> <td>Open Ground</td> <td>5000</td> <td>Campus Park</td> </tr> <tr> <td>*</td> <td></td> <td></td> <td></td> </tr> </tbody> </table>			Venue Name	Capacity	Location	▶	Main Auditorium	1000	Block A		Seminar Hall	300	Block B		Open Ground	5000	Campus Park	*			
	Venue Name	Capacity	Location																		
▶	Main Auditorium	1000	Block A																		
	Seminar Hall	300	Block B																		
	Open Ground	5000	Campus Park																		
*																					

UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
AddBtn	Button	VenuesUC.cs	<code>\$@"INSERT INTO venues(venue_name, capacity, location) VALUES ('{venue.VenueName}', {venue.Capacity}, '{venue.Location}')"</code>
UpdateBtn	Button	VenuesUC.cs	<code>\$@"UPDATE venues SET venue_name = '{venue.VenueName}', capacity = {venue.Capacity}, location = '{venue.Location}' WHERE venue_id = {venue.VenueId}"</code>
DeleteBtn	Button	VenuesUC.cs	<code>\$"DELETE FROM venues WHERE venue_id = {venueId}"</code>
DataGridView1	DataGridView	VenuesUC.cs	Venues Table

Interface Name	VenueAllocationUC
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Venues

Venues

Venue Allocations

Event:

Venue:

Assigned_at:

Assigned(time):

	Event	Venue	Assigned_date	Assigned_time
▶	AI Hackathon	Main Auditorium	15/03/2023	11:30:00
	Tech Summit	Seminar Hall	15/03/2023	12:30:00
	new	Open Ground	17/03/2023	12:30:00
*				

UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
AddBtn	Button	VenueAllocationUC.cs	<code>\$@"INSERT INTO venue_allocations(event_id, venue_id, assigned_date, assigned_time) VALUES ({allocation.EventId}, {allocation.VenueId},{allocation.AssignedDate:yyyy-MM-dd}','{allocation.AssignedTime:hh\\:mm\\:ss}')</code> "
UpdateBtn	Button	VenueAllocationUC.cs	<code>\$@"UPDATE venue_allocations SET event_id = {allocation.EventId}, venue_id = {allocation.VenueId}, assigned_date = '{allocation.AssignedDate: yyyy-MM-dd}', assigned_time = '{allocation.AssignedTime:hh\\:mm\\:ss}' WHERE venue_allocation_id = {allocation.VenueAllocationId}"</code>
DeleteBtn	Button	VenueAllocationUC.cs	<code>\$"DELETE FROM venue_allocations WHERE venue_allocation_id = {allocationId}"</code>
cmbEvent	Combo Box	VenueAllocationUC.cs	<code>"SELECT event_id, event_name FROM itec_events"</code>
cmbVenue	Combo Box	VenueAllocationUC.cs	<code>"SELECT venue_id, venue_name FROM venues"</code>
DataGridView1	DataGridView	VenueAllocationUC.cs	<code>"SELECT va.venue_allocation_id,va.event_id, va.venue_id, va.assigned_date, va.assigned_time, v.venue_name, ie.event_name FROM venue_allocations va LEFT JOIN venues v on v.venue_id=va.venue_id LEFT JOIN itec_events ie on ie.event_id=va.event_id"</code>

Interface Name

CommitteeMembersUC

Committee_Role Management

Committee Members

Committees

Member Name:

Committee:

Technical Committ

Role:

Insert

Update

Delete

	Committee	Name	Role
▶	Technical Committee	Ali	Faculty Advisor
	Technical Committee	Ahmed	Student Lead
*			

UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
AddBtn	Button	CommitteeMembersUC.cs	<pre> \$@"INSERT INTO committee_members (committee_id, name, role_id) VALUES ({member.CommitteeId}, '{member.Name}', {(member.RoleId.HasValue ? member.RoleId.ToString() : "NULL")})" </pre>
UpdateBtn	Button	CommitteeMembersUC.cs	<pre> \$@"UPDATE committee_members SET committee_id = {member.CommitteeId}, name = '{member.Name}', role_id = {(member.RoleId.HasValue ? member.RoleId.ToString() : "NULL")} WHERE member_id = {member.MemberId}" </pre>
DeleteBtn	Button	CommitteeMembersUC.cs	<pre> \$"DELETE FROM committee_members WHERE member_id = {memberId}" </pre>
cmbCommittee	Combo Box	CommitteeMembersUC.cs	<pre> "SELECT event_id, event_name FROM itec_events" </pre>
cmbRole	Combo Box	CommitteeMembersUC.cs	<pre> "SELECT participant_id, name FROM participants" </pre>

DataGridView1	DataGridView	EventsResultsUC.cs	<pre>@"SELECT cm.member_id, cm.committee_id, cm.name, cm.role_id,c.committee_name, l.value AS role_name FROM committee_members cm LEFT JOIN committees c on cm.committee_id = c.committee_id LEFT JOIN lookup l on cm.role_id = l.lookup_id" e.event_name, p.name AS participant_name, t.team_name FROM event_results er LEFT JOIN itec_events e on er.event_id = e.event_id LEFT JOIN participants p on er.participant_id = p.participant_id LEFT JOIN teams t on er.team_id = t.team_id"</pre>
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Interface Name

CommitteesUC

Committee Role Management

Committee Members

Committees

Itec:

2021

Committee Name:

Insert

Delete

Update

	Year	CommitteeName
▶	2023	Technical Committee
	2023	Logistics Committee
	2024	Security Task Force
	2025	Quantum Planning Committee
*		

UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
AddBtn	Button	CommitteesUC.cs	<pre> \$@"INSERT INTO committees (itec_id, committee_name) VALUES ({newCommittee.ItecId}, '{newCommittee.CommitteeName}')" </pre>

UpdateBtn	Button	CommitteesUC.cs	\$@"UPDATE committees SET itec_id={updatedCommittee.ItecId}, committee_name = '{updatedCommittee.CommitteeName}' WHERE committee_id = {updatedCommittee.CommitteeId}"
DeleteBtn	Button	CommitteesUC.cs	\$"DELETE FROM committees WHERE committee_id = {committeeId}"
cmbItec	Combo Box	CommitteesUC.cs	"SELECT itec_id, year FROM itec_editions"
DataGridView1	DataGridView	CommitteesUC.cs	"SELECT c.committee_id,c.itec_id,ie.year,c.committee_name FROM committees c LEFT JOIN itec_editions ie on ie.itec_id=c.itec_id"

Interface Name	FinancesUC																																																																																																			
Financial Management Sponsorship Tracking Sponsors Finances Management																																																																																																				
Itec: 2021 Event: sd Type: Sponsorship Inco Amount: 0 From Type: From ID: To Type: TO ID: Date: Sunday , 23 March 2025 Description: Insert Delete Update	<table border="1" style="width: 100%; border-collapse: collapse; text-align: center;"> <thead> <tr style="background-color: #d9d9d9;"> <th></th> <th>ItecYear</th> <th>EventName</th> <th>FinanceType</th> <th>Amount</th> <th>From</th> <th>To</th> <th>Description</th> <th>Date</th> </tr> </thead> <tbody> <tr> <td>▶</td> <td>2023</td> <td>sd</td> <td>Sponsorship ...</td> <td>Rs750,000.00</td> <td>Sponsor: Tech ...</td> <td>Committee: Tec...</td> <td>Platinum Spons...</td> <td>2025-03-22</td> </tr> <tr> <td></td> <td>2023</td> <td>sdf</td> <td>Expense</td> <td>Rs250,000.00</td> <td>Vendor: Lahore ...</td> <td>Committee: Tec...</td> <td>Venue Setup & ...</td> <td>2025-03-22</td> </tr> <tr> <td></td> <td>2023</td> <td>AI Hackathon</td> <td>Expense</td> <td>Rs500,000.00</td> <td>Sponsor: Tech ...</td> <td>Committee: Se...</td> <td>Platinum Spons...</td> <td>2025-03-22</td> </tr> <tr> <td></td> <td>2023</td> <td>Tech Summit</td> <td>Ticket Sales</td> <td>Rs300,000.00</td> <td>Sponsor: Future...</td> <td>Committee: Se...</td> <td>Gold Sponsorsh...</td> <td>2025-03-22</td> </tr> <tr> <td></td> <td>2023</td> <td>new</td> <td>Expense</td> <td>Rs150,000.00</td> <td>Vendor: Lahore ...</td> <td>Committee: Qu...</td> <td>Food Services</td> <td>2025-03-22</td> </tr> <tr> <td></td> <td>2023</td> <td>AI Hackathon</td> <td>Expense</td> <td>Rs75,000.00</td> <td>Vendor: Event P...</td> <td>Committee: Se...</td> <td>AV Equipment ...</td> <td>2025-03-22</td> </tr> <tr> <td></td> <td>2023</td> <td>AI Hackathon</td> <td>Ticket Sales</td> <td>Rs120,000.00</td> <td>User: shahzaib</td> <td>Committee: Se...</td> <td>Hackathon Reg...</td> <td>2025-03-22</td> </tr> <tr> <td></td> <td>2023</td> <td>sd</td> <td>Ticket Sales</td> <td>Rs45,000.00</td> <td>User: new</td> <td>Committee: Qu...</td> <td>Seminar Tickets</td> <td>2025-03-22</td> </tr> <tr> <td></td> <td>2021</td> <td>AI Hackathon</td> <td>Ticket Sales</td> <td>Rs100.00</td> <td>Sponsor:</td> <td>Sponsor:</td> <td>jasn</td> <td>2025-03-23</td> </tr> <tr> <td>*</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </tbody> </table> Generate Report: Income: Rs0.00 Expenses: Rs0.00 Balance: Rs0.00		ItecYear	EventName	FinanceType	Amount	From	To	Description	Date	▶	2023	sd	Sponsorship ...	Rs750,000.00	Sponsor: Tech ...	Committee: Tec...	Platinum Spons...	2025-03-22		2023	sdf	Expense	Rs250,000.00	Vendor: Lahore ...	Committee: Tec...	Venue Setup & ...	2025-03-22		2023	AI Hackathon	Expense	Rs500,000.00	Sponsor: Tech ...	Committee: Se...	Platinum Spons...	2025-03-22		2023	Tech Summit	Ticket Sales	Rs300,000.00	Sponsor: Future...	Committee: Se...	Gold Sponsorsh...	2025-03-22		2023	new	Expense	Rs150,000.00	Vendor: Lahore ...	Committee: Qu...	Food Services	2025-03-22		2023	AI Hackathon	Expense	Rs75,000.00	Vendor: Event P...	Committee: Se...	AV Equipment ...	2025-03-22		2023	AI Hackathon	Ticket Sales	Rs120,000.00	User: shahzaib	Committee: Se...	Hackathon Reg...	2025-03-22		2023	sd	Ticket Sales	Rs45,000.00	User: new	Committee: Qu...	Seminar Tickets	2025-03-22		2021	AI Hackathon	Ticket Sales	Rs100.00	Sponsor:	Sponsor:	jasn	2025-03-23	*								
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UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
AddBtn	Button	FinancesUC.cs	<pre> \$@"INSERT INTO finances(itec_id, event_id, type_id, amount, from_entity_type, from_entity_id, to_entity_type, to_entity_id, description, date_recorded) VALUES ({transaction.ITECId}, {eventPart}, {transaction.TypeId}, {transaction.Amount}, '{transaction.FromEntityType}', {transaction.FromEntityId}, '{transaction.ToEntityType}', {transaction.ToEntityId}, '{transaction.Description}', '{transaction.DateRecorded:yyyy-MM-dd HH:mm:ss}')" </pre>
UpdateBtn	Button	FinancesUC.cs	<pre> \$@"UPDATE finances SET itec_id = {transaction.ITECId}, event_id = {eventPart}, type_id = {transaction.TypeId}, amount = {transaction.Amount}, from_entity_type = '{transaction.FromEntityType}', from_entity_id = {transaction.FromEntityId}, to_entity_type = '{transaction.ToEntityType}', to_entity_id = {transaction.ToEntityId}, description = '{transaction.Description}', date_recorded = '{transaction.DateRecorded:yyyy-MM-dd HH:mm:ss}' WHERE transaction_id = {transaction.TransactionId}" </pre>
DeleteBtn	Button	FinancesUC.cs	<pre> \$"DELETE FROM finances WHERE transaction_id = {transactionId}" </pre>
cmbEvent	Combo Box	FinancesUC.cs	<pre> "SELECT event_id, event_name FROM itec_events" </pre>
cmbItec	Combo Box	FinancesUC.cs	<pre> "SELECT event_id, event_name FROM itec_events" </pre>
cmbType	Combo Box	FinancesUC.cs	<pre> "SELECT lookup_id, value FROM lookup WHERE category = 'FinanceType'" </pre>
DataGridView1	DataGridView	FinancesUC.cs	<pre> @"SELECT f.transaction_id, f.itec_id, f.event_id, f.type_id, f.amount, f.from_entity_type, f.from_entity_id, f.to_entity_type, f.to_entity_id, f.description, f.date_recorded, l.value AS finance_type, e.event_name, ie.year AS itec_year, COALESCE(s.sponsor_name, u.username, </pre>

			c.committee_name, v.vendor_name) AS from_entity_name, COALESCE(s2.sponsor_name, u2.username, c2.committee_name, v2.vendor_name) AS to_entity_name FROM finances f LEFT JOIN lookup l ON f.type_id = l.lookup_id LEFT JOIN itec_events e on f.event_id = e.event_id LEFT JOIN itec_editions ie ON f.itec_id = ie.itec_id LEFT JOIN sponsors s ON f.from_entity_type = 'Sponsor' AND f.from_entity_id = s.sponsor_id LEFT JOIN users u ON f.from_entity_type = 'User' AND f.from_entity_id = u.user_id LEFT JOIN committees c ON f.from_entity_type = 'Committee' and f.from_entity_id = c.committee_id LEFT JOIN vendors v on f.from_entity_type = 'Vendor' AND f.from_entity_id = v.vendor_id LEFT JOIN sponsors s2 ON f.to_entity_type = 'Sponsor' AND f.to_entity_id = s2.sponsor_id LEFT JOIN users u2 on f.to_entity_type = 'User' and f.to_entity_id = u2.user_id LEFT JOIN committees c2 on f.to_entity_type = 'Committee' AND f.to_entity_id = c2.committee_id LEFT JOIN vendors v2 ON f.to_entity_type = 'Vendor' and f.to_entity_id = v2.vendor_id"
GenerateReportBtn	Button	FinancesUC.cs	\$@"SELECT SUM(CASE WHEN l.value IN ('Sponsorship Income', 'Ticket Sales') THEN f.amount ELSE 0 END) as TotalIncome, SUM(CASE WHEN l.value = 'Expense' THEN f.amount ELSE 0 END) AS TotalExpenses, (SUM(CASE WHEN l.value IN ('Sponsorship', 'Ticket Sales') then f.amount ELSE 0 END) - SUM(CASE WHEN l.value = 'Expense' THEN f.amount ELSE 0 END)) AS NetBalance FROM finances f JOIN lookup l ON f.type_id = l.lookup_id WHERE f.itec_id = {itecId}"

Interface Name

SponsorshipUC

Financial Management

Sponsorship Tracking

Sponsors

Finances Management

Sponsor Name:

Contact:

Insert

Update

Delete

	Sponsor Name	Contact	TotalAmount
▶	Tech Giants Inc	sponsor@techgiants.com	1250000.00
	Future Ventures	contact@futureventures....	300000.00
	Innovate Corp	partner@innovatecorp.org	0.00
	PTCL		0.00
	Jazz		0.00
	Systems Limited		0.00
*			

UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
AddBtn	Button	SponsorshipUC.cs	\$@"INSERT INTO sponsors (sponsor_name, contact) VALUES ('{newSponsor.SponsorName}', '{newSponsor.Contact}')"
UpdateBtn	Button	SponsorshipUC.cs	\$@"UPDATE sponsors SET sponsor_name = '{updatedSponsor.SponsorName}', contact = '{updatedSponsor.Contact}' WHERE sponsor_id = {updatedSponsor.SponsorId}"
DeleteBtn	Button	SponsorshipUC.cs	\$"DELETE FROM sponsors WHERE sponsor_id = {sponsorId}"

DataGridView1	DataGridView	SponsorshipUC.cs	@ "SELECT s.sponsor_id, s.sponsor_name, s.contact, COALESCE(SUM(f.amount), 0) AS total_amount FROM sponsors s LEFT JOIN finances f ON s.sponsor_id = f.from_entity_id AND f.from_entity_type = 'Sponsor' GROUP BY s.sponsor_id, s.sponsor_name, s.contact"
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Interface Name	DashboardUC
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Financial Management
Sponsorship Tracking
Sponsors
Finances Management

Sponsor Name:

Contact:

Insert
Update
Delete

	Sponsor Name	Contact	TotalAmount
▶	Tech Giants Inc	sponsor@techgiants.com	1250000.00
	Future Ventures	contact@futureventures....	300000.00
	Innovate Corp	partner@innovatecorp.org	0.00
	PTCL		0.00
	Jazz		0.00
	Systems Limited		0.00
*			

UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
ApproveBtn	Button	DashboardUC.cs	\$"UPDATE users SET role_id = {roleId} WHERE user_id = {userId}"
cmbRole	Button	DashboardUC.cs	"SELECT role_id, role_name FROM roles"
DataGridView1	DataGridView	DashboardUC.cs	"SELECT * FROM users WHERE role_id IS NULL"

Interface Name	DutiesUC
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Duty Assignment _Tracking

Committee:

Assigned to:

Deadline:

Status:

Task Description:

	CommitteeName	Assigned_to	Task_Description	Deadline	Status
▶	Security Task Fo...	asjjasn	s	2025-03-22	Pending
*					

UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
AddBtn	Button	DutiesUC.cs	\$@"INSERT INTO duties (committee_id, assigned_to, task_description, deadline, status_id) VALUES ({newDuty.CommitteeId}, '{newDuty.AssignedTo}', '{newDuty.TaskDescription}', '{newDuty.Deadline:yyyy-MM-dd}', {newDuty.StatusId})"
UpdateBtn	Button	DutiesUC.cs	\$@"UPDATE duties SET committee_id = {updatedDuty.CommitteeId}, assigned_to = '{updatedDuty.AssignedTo}', task_description = '{updatedDuty.TaskDescription}', deadline = '{updatedDuty.Deadline:yyyy-MM-dd}', status_id = {updatedDuty.StatusId} WHERE duty_id = {updatedDuty.DutyId}"

DeleteBtn	Button	DutiesUC.cs	\$"DELETE FROM duties WHERE duty_id = {dutyId}"
cmbCommittee	Combo Box	DutiesUC.cs	"SELECT committee_id, committee_name FROM committees"
cmbStatus	Combo Box	DutiesUC.cs	"SELECT lookup_id, value FROM lookup WHERE category = 'DutyStatus'"
DataGridView1	DataGridView	DutiesUC.cs	@ "SELECT d.duty_id,d.committee_id, d.assigned_to,d.task_description,d.deadline ,d.status_id,c.committee_name,l.value AS status_name FROM duties d LEFT JOIN committees c on d.committee_id = c.committee_id LEFT JOIN lookup l on d.status_id = l.lookup_id"

Interface Name	EditioncUC
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ITEC Editions Management

Year:

Theme:

Description:

	Year	Theme	Description	NoOfEvents	NoOfParticipants
▶	2021	Digital Frontier	Embracing digital transfor...	0	4
	2022	Green Tech	Sustainable technology sol...	0	1
	2023	AI Revolution	Advanced AI applications	3	1
	2024	Cyber Security	Next-gen security solutions	2	0
	2025	Quantum Com...	Future of computing techn...	0	0
*					

UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
AddBtn	Button	EditioncUC.cs	\$@"INSERT INTO itec_editions(year, theme, description) VALUES({newEdition.Year}, '{newEdition.Theme}', '{newEdition.Description}')"

UpdateBtn	Button	EditioncUC.cs	\$@"UPDATE itec_editions SET year = {updatedEdition.Year}, theme = '{updatedEdition.Theme}',description = '{updatedEdition.Description}' WHERE itec_id = {updatedEdition.ItecId}"
DeleteBtn	Button	EditioncUC.cs	\$"DELETE FROM itec_editions WHERE itec_id = {itecId}"
DataGridView1	DataGridView	EditioncUC.cs	@ "SELECT ie.itec_id, ie.year, ie.theme, ie.description,COUNT(distinct ievent.event_id) AS event_count, COUNT(distinct p.participant_id) AS participant_count FROM itec_editions ie LEFT JOIN itec_events ievent on ie.itec_id = ievent.itec_id LEFT JOIN participants p on ie.itec_id = p.itec_id GROUP BY ie.itec_id, ie.year, ie.theme, ie.description"

Business Report-1

Business Report Name: **Financial Report**

Sample of report:

FULL FINANCIAL REPORT

2023

TransactionType	Amount	Date	From	To	Description
Sponsorship Income	750000.00	2025-03-22	Tech Giants Inc.	Technical Committee	Platinum Sponsorship - Jazz
Expense	250000.00	2025-03-22	Lahore Catering	Technical Committee	Venue Setup & Logistics
Expense	500000.00	2025-03-22	Tech Giants Inc.	Security Task Force	Platinum Sponsorship
Ticket Sales	300000.00	2025-03-22	Future Ventures	Security Task Force	Gold Sponsorship
Expense	150000.00	2025-03-22	Lahore Catering	Quantum Planning Committee	Food Services
Expense	75000.00	2025-03-22	Event Pro	Security Task Force	AV Equipment Rental
Ticket Sales	120000.00	2025-03-22	shahzaib	Security Task Force	Hackathon Registrations
Ticket Sales	45000.00	2025-03-22	new	Quantum Planning Committee	Seminar Tickets

Income Breakdown:

Sponsorship Income: 750,000.00

Ticket Sales: 465,000.00

Expense Breakdown:

Expense: 975,000.00

Total Income:	1,215,000.00
Total Expenses:	975,000.00
Net Balance:	240,000.00

2021

TransactionType	Amount	Date	From	To	Description
Ticket Sales	100.00	2025-03-23			jasn

Income Breakdown:

Ticket Sales: 100.00

Expense Breakdown:

Total Income:	100.00
Total Expenses:	0.00
Net Balance:	100.00

Query to build Report:

```
@ " SELECT e.year AS Year, l.value AS TransactionType, f.amount AS Amount,
      DATE_FORMAT(f.date_recorded, '%Y-%m-%d') AS TransactionDate,
      COALESCE(s.sponsor_name, v.vendor_name, u.username, c.committee_name) AS FromEntity,
      COALESCE(s2.sponsor_name, v2.vendor_name, u2.username, c2.committee_name) AS ToEntity,
      f.description AS Description,
CASE
  WHEN l.value IN ('Sponsorship Income', 'Ticket Sales', 'Miscellaneous Income') THEN 'Income'
  WHEN l.value IN ('Vendor Payment', 'Expense', 'Venue Rental Expense', 'Equipment Purchase',
'Marketing Expense') THEN 'Expense'
END AS FinancialCategory
FROM finances f
JOIN itec_editions e ON f.itec_id = e.itec_id
JOIN lookup l ON f.type_id = l.lookup_id
```

```

LEFT JOIN sponsors s ON f.from_entity_type = 'Sponsor' AND f.from_entity_id = s.sponsor_id
LEFT JOIN vendors v ON f.from_entity_type = 'Vendor' AND f.from_entity_id = v.vendor_id
LEFT JOIN users u ON f.from_entity_type = 'User' AND f.from_entity_id = u.user_id
LEFT JOIN committees c ON f.from_entity_type = 'Committee' AND f.from_entity_id = c.committee_id
LEFT JOIN sponsors s2 ON f.to_entity_type = 'Sponsor' AND f.to_entity_id = s2.sponsor_id
LEFT JOIN vendors v2 ON f.to_entity_type = 'Vendor' AND f.to_entity_id = v2.vendor_id
LEFT JOIN users u2 ON f.to_entity_type = 'User' AND f.to_entity_id = u2.user_id
LEFT JOIN committees c2 ON f.to_entity_type = 'Committee' AND f.to_entity_id = c2.committee_id"

```

Business Report-2

Business Report Name: **Venue Report**

Sample of report:

FULL VENUE REPORT

Main Auditorium

Year	Venue	Capacity	Date	Time	Event
2023	Main Auditorium	1000	2023-03-15	11:30	AI Hackathon

Seminar Hall

Year	Venue	Capacity	Date	Time	Event
2023	Seminar Hall	300	2023-03-15	12:30	Tech Summit

Open Ground

Year	Venue	Capacity	Date	Time	Event
2024	Open Ground	5000	2023-03-17	12:30	new

Query to build Report:

```

@" SELECT e.year AS Year, v.venue_name AS Venue, v.capacity AS Capacity,
    ev.event_name AS AssignedEvent,
    DATE_FORMAT(va.assigned_date, '%Y-%m-%d') AS AssignmentDate,
    TIME_FORMAT(va.assigned_time, '%H:%i') AS AssignmentTime
FROM venue_allocations va
JOIN venues v ON va.venue_id = v.venue_id
JOIN itec_events ev ON va.event_id = ev.event_id
JOIN itec_editions e ON ev.itec_id = e.itec_id";

```

Business Report-3

Business Report Name: **Event Report**

Sample of report:

FULL EVENT REPORT

sd

Event	Category	Date	Committee	Venue	Participants
sd	Seminar	2023-07-24	Logistics Committee	Main Auditorium	1

sdf

Event	Category	Date	Committee	Venue	Participants
sdf	Competition	2023-07-24	Technical Committee	Seminar Hall	0

AI Hackathon

Event	Category	Date	Committee	Venue	Participants
AI Hackathon	Competition	2023-03-15	Technical Committee	Main Auditorium	2

Tech Summit

Event	Category	Date	Committee	Venue	Participants
Tech Summit	Seminar	2023-03-18	Logistics Committee	Seminar Hall	1

new

Event	Category	Date	Committee	Venue	Participants
new	Exhibition	2025-07-24	Technical Committee	Seminar Hall	0

Query to build Report:

```
@ "SELECT e.year AS Year, ev.event_name AS Event, ec.category_name AS Category,
DATE_FORMAT(ev.event_date, '%Y-%m-%d') AS EventDate,
c.committee_name AS ResponsibleCommittee, v.venue_name AS Venue,
COUNT(ep.participant_id) AS ParticipantsRegistered
FROM itec_events ev
JOIN itec_editions e ON ev.itec_id = e.itec_id
LEFT JOIN event_categories ec ON ev.event_category_id = ec.event_category_id
LEFT JOIN committees c ON ev.committee_id = c.committee_id
LEFT JOIN venues v on ev.venue_id = v.venue_id
LEFT JOIN event_participants ep ON ev.event_id = ep.event_id
GROUP BY ev.event_id, e.year, ev.event_name, ec.category_name, ev.event_date, c.committee_name,
v.venue_name";
```